

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

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Abstract

Long-horizon video generation is evaluated with whole-frame metrics that reward motion and temporal consistency. For fixed-camera nature scenes this creates an ambiguity: motion of water, fire, smoke, or rain is desirable, whereas motion of the background is an error. A system can therefore score well on motion while its scene drifts, or on consistency while its flow stagnates. We introduce SNF-Bench, an evaluation framework for long-horizon fixed-camera generation that partitions each scene into static support and dynamic flow and reports static fidelity, flow persistence with absolute magnitude, and drift leakage separately, never as one score. Drift leakage is interpretive context rather than a headline measurement. Each factor is validated mechanistically rather than by correlation with preference: we inject global translation, rotation, and scale drift and progressive motion attenuation at known severity into real generations, and require each factor to respond in its stated direction and to remain selective against corruptions it does not target. Auditing publicly released long-horizon text-conditioned checkpoints under one recorded common inference configuration, plus an image-conditioned track with released-pipeline references and a deployment-sensitivity panel, we find that whole-frame motion and static-region drift induce near-opposite orderings of the same outputs. At maximum controlled translation, fBD and NBF rise to $1.86\times$ and $1.32\times$ baseline, but whole-frame Dynamic Degree reaches only $1.07\times$ —rewarding the corruption. SNF-Bench measures where motion occurs and whether it persists; it does not measure physical realism.

1. Introduction

Long-horizon, autoregressive (AR), and streaming video diffusion models now extend generation from sec-

onds to minute-scale horizons [20, 32, 34, 36, 38, 39]. Evaluation has not kept pace: small temporal errors invisible in short clips accumulate over long rollouts into background displacement, color drift, or motion decay, and whole-frame metrics summarize these qualitatively different failures into one score.

The problem is sharpest in fixed-camera nature scenes. In a static-camera video of a river, the water should move while the bank, rocks, and surrounding landscape stay fixed. The same separation governs waterfalls, ocean waves, rainfall, fire, smoke, and wind-driven vegetation: some regions form static support that must remain structurally fixed, while others carry dynamic flow that must persist. Whole-frame motion measures reward optical flow regardless of its origin, crediting background drift as readily as genuine fluid motion, while temporal-consistency measures can favor videos whose stability comes from suppressing the flow they should sustain. Two opposite failures—static drift and flow decay—are thereby conflated, and increasingly so at long horizons, where accumulated background motion can dominate a sequence’s apparent dynamics. Figure 1 shows both failures in outputs of released checkpoints under the common setting.

We introduce SNF-Bench (Static-Natural Flow), an evaluation framework built around this decomposition, evaluated at 60 and 120s horizons with shorter (5s) and longer (240s) tiers retained as diagnostics. Measurement is organized into three factor groups, the third interpretive. Static fidelity captures geometric displacement and residual flow inside nominally stationary regions. Flow persistence reports the retention of dynamic-region motion with its absolute compensated magnitude, so a ratio cannot look strong merely because motion is uniformly weak. Drift leakage relates static-region motion and global-motion compensation to dynamic-region flow; it contextualizes the two measurement axes rather than adding a third, and the central claim does not rest on it.

A benchmark is only as trustworthy as its measure-

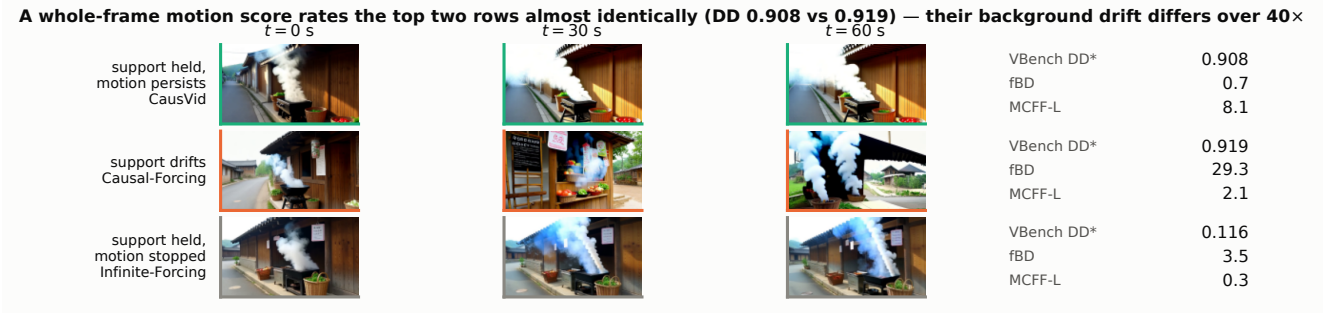


Figure 1. Why a fixed-camera benchmark needs a spatial decomposition. Three 60s outputs from released checkpoints under the recorded common configuration. A whole-frame motion score rates the first two rows almost identically, yet only the first holds its scene in place; their static-region drift differs by over an order of magnitude. The third row shows the opposite failure, which that score does detect. DD* is the per-video quantity underlying VBench [11] Dynamic Degree, computed under its published rule but read before the binarization the suite applies when aggregating, so that near-identical rows remain distinguishable as numbers.

ments, so we validate the protocol independently of any model comparison. Controlled perturbations inject global translation, rotation, and scale drift, progressive motion attenuation, late freezing, photometric drift, and temporal repetition into real fixed-camera generations at known severity, and we require each factor to respond in the direction its definition states and to remain selective against corruptions it does not target. Because a parameter expressed in degrees is not commensurate with one expressed in pixels, geometric severity is specified as the mean displacement a corruption induces, which makes the families directly comparable. We additionally recompute every measurement under systematic erosion and dilation of the region partition.

We then audit publicly released long-horizon generators as executed through the recorded evaluation pipeline. The audit exposes cases where conventional metrics and SNF-Bench interpret the same outputs differently: high whole-frame motion coexisting with substantial static-region displacement, and high stability coexisting with decaying flow. We do not argue that existing metrics are incorrect. They answer a whole-frame question accurately; that question does not encode the spatial distinction this setting requires. Nor does SNF-Bench measure physical realism—a sequence may be drift-free and persistent yet physically implausible. The contribution is narrower and complementary: making the spatial origin and temporal persistence of motion explicit.

Our contributions are:

1. SNF-Bench, an evaluation framework for long-horizon fixed-camera generation organized around a static-support/dynamic-flow partition.
2. A measurement protocol that reports static fidelity,

flow persistence with absolute magnitude, and drift leakage separately, never as a single score, the last as an interpretive diagnostic.

3. Mechanistic validation of every reported factor under corruptions of known type and severity, with selectivity measured against the families each factor should ignore and every measurement recomputed under erosion and dilation of the region partition.
4. A public-model audit showing that this decomposition materially changes the interpretation and ordering of long-horizon behavior relative to whole-frame metrics.

2. Related Work

Video-generation evaluation. General-purpose suites assess visual quality, temporal consistency, motion, and semantic alignment. VBench [11] decomposes quality into dimensions including background consistency, motion smoothness and dynamic degree, and VBench-2.0 [41] extends it toward intrinsic faithfulness; EvalCrafter [21] learns a weighting of many metrics from user opinion; Video-Bench [6] scores generations with multimodal language models validated against human preference. Others specialize: T2V-CompBench [28] targets compositional binding, VM-Bench [19] perceptually aligned motion quality, and WorldScore [3] controllability and dynamics for world generation. These set the standards a benchmark paper must meet—explicit taxonomy, validated metrics, stated scale—but their measurements are whole-frame or sequence-level, and such an aggregate cannot reveal whether motion originates where intended or in displacement of nominally static structure. VM-Bench is the closest comparator, but it asks whether motion is perceptually plausible where we ask where

it occurs and whether it survives compensation for global displacement. Our validation is correspondingly mechanistic—perturb by a known amount, require the expected response—complementing the human-alignment evidence those suites provide.

Long-horizon autoregressive generation. Autoregressive and streaming generators [2, 13, 18, 29, 42] extend synthesis beyond short bidirectional clips by producing frames or temporal chunks causally, including CausVid [37], Self-Forcing [10], Infinite-Forcing [12], Rolling-Forcing [20], Reward-Forcing [22], LongLive [34], and Causal-Forcing [43] with its few-step successor Causal-Forcing++ [40]; bidirectional diffusion systems [1, 5, 14, 31, 35] provide short-horizon reference points. These systems differ substantially in temporal block size, training objective, memory and cache policy, and horizon-extension mechanism [36], yet their evaluation typically relies on generic video metrics or method-specific qualitative analysis. That diversity motivates a recorded common-configuration audit, and the evaluation gap motivates a system-agnostic question: does static support remain fixed while dynamic flow persists?

Image dynamics and nature-flow animation. Work on animating still images has long modeled natural motion explicitly [9, 15, 17, 23, 33]: controllable fluid animation separates static image structure from specified fluid motion [24], and Generative Image Dynamics models long-term scene motion through learned trajectory representations [16]. These works treat static support and dynamic content as distinct modeling targets; SNF-Bench elevates that distinction to an evaluation principle for long-horizon generators.

3. SNF-Bench Design

SNF-Bench evaluates long-horizon generation under a deliberately constrained setting: a fixed camera observing a scene containing both stationary structure and persistent natural motion. This isolates two failure modes that are difficult to separate in unconstrained video—static drift, where nominally stationary structure moves over time, and flow decay, where intended dynamics weaken or disappear. Every sequence is treated as a composition of two spatial roles, Ω_{static} for regions expected to remain geometrically stationary and Ω_{flow} for regions expected to exhibit sustained motion. The partition defines an evaluation role, not a semantic ontology: a riverbank is static support while the river surface is flow; buildings are static support in a rainfall scene while the falling rain is flow. SNF-Bench does not require dynamics inside Ω_{flow} to follow

any physical model. It measures whether motion stays localized to the appropriate regions and persists over time without mixing static support. Every system receives identical conditioning per item, and no prompt is rewritten for a particular method. Both conditioning modalities are evaluated under the same decomposition, differing only in how static support is established: T2V builds it in the opening sequence, while I2V takes it from a shared source image, so divergence there is attributable to the system rather than to a differently imagined scene. The main paper reports T2V at 60/120 s and the I2V deployment panel at 60 s; all four 5/60/120/240 s tiers are supplementary.

3.1. Scene Categories and Horizons

Prompts are stratified across representative fixed-camera natural processes—channel water, ocean waves, precipitation, fire and smoke, lava, and wind-borne motion—with the category set fixed before the final sweep. The realized composition is dominated by directional and broad-area water motion with additional atmospheric and combustion categories, and conclusions should be read at that scope. The strata are not equally populated: channel water accounts for 61% of the principal prompt set. Reported aggregates are means over prompts, not category macro-averages: three of the five populated categories hold two prompts each, so equal category weighting would buy balance at a cost in variance the sample cannot support. We instead recompute every system both ways and state which conclusions survive, with complete per-category counts in the supplementary.

Four temporal regimes are used: 5 s as an initialization check, 60 s as the primary reported horizon, 120 s as a supporting tier, and 240 s as a diagnostic extreme. Conclusions rest on the 60 s tier; the 120 s tier is reported alongside it as a supporting check, and the 5 and 240 s tiers are in the supplementary material. A run that cannot reach a target horizon under its declared track and setting is reported as unsupported rather than silently omitted.

3.2. Region Partition

The partition is obtained automatically and identically for every sequence. Mean optical-flow magnitude over the leading 12% of sampled frame pairs is thresholded by Otsu’s method [25] into a dynamic region and its complement; the static region is then eroded and a 4% frame border discarded, which together form an ignored transition band so that boundary pixels—where the partition is least certain and flow least reliable—contribute to neither region. Restricting the estimate to an early window reduces circularity from late ac-

246 cumulated drift: at 60s the partition is fixed from
247 the first 7.1s and is therefore not defined by the late-
248 window failure being measured. It still depends on
249 early checkpoint behavior and on the same flow esti-
250 mator used downstream. Two limitations bound the
251 later conclusions. The partition is automatic, so a sys-
252 tem drifting within the first 12% of its rollout can in-
253 fluence its own partition; we therefore treat partitions
254 as measured quantities and recompute every result un-
255 der systematic erosion and dilation of the boundary
256 (Sec. 5). And a two-way partition cannot represent
257 layered content: rain and snow transit static support,
258 so those pixels are assigned by flow magnitude alone
259 and static-fidelity values in precipitation scenes carry
260 an unmodeled contribution. Derivation, the sensitiv-
261 ity study, and the annotation protocol we regard as
262 the natural strengthening of this component are in the
263 supplementary material.

264 3.3. Benchmark Manifest

265 SNF-Bench is defined by a versioned manifest fixed
266 before the final sweep, recording per item: identifier
267 and category, exact prompt, target horizons, resolu-
268 tion and frame-rate metadata, and provenance. Af-
269 ter freezing, items are never removed because a sys-
270 tem performs poorly; corrections require a new bench-
271 mark version. A failure taxonomy—unsupported hori-
272 zon, out-of-memory, runtime crash, numerical corrup-
273 tion, invalid output—is likewise fixed in advance, and
274 failures are reported alongside successful generations
275 rather than silently excluded. Coverage is counted
276 from valid per-video measurements rather than from
277 the presence of a results file, so a run that terminated
278 early is reported as partial.

279 4. Drift–Motion Evaluation Protocol

280 SNF-Bench evaluates fixed-camera generation along
281 three factor groups: Static Fidelity and Flow Persis-
282 tence are the measurement axes, and Drift Leakage
283 (DLR, DAR) contextualizes them rather than adding
284 a third measurement. No individual score is sufficient:
285 low background motion can be achieved by freezing the
286 entire video, and high foreground motion can be pro-
287 duced by drifting the entire scene. We therefore always
288 report static-region stability jointly with both the re-
289 lative persistence and absolute magnitude of dynamic-
290 region motion. Let I_t denote frame t , Ω_{static} and Ω_{flow}
291 the region masks of Sec. 3.2, and $\mathbf{u}_t$ the optical flow
292 from I_t to I_{t+1} , estimated with RAFT [30]. Each se-
293 quence defines an early window $\mathcal{E}$ and a late window $\mathcal{L}$
294 of equal duration, $\mathcal{E}$ beginning after a warm-up so that
295 generation transients do not define the motion refer-
296 ence; extents are fixed in the specification.

4.1. Static Fidelity

297 Feature-aligned Background Drift (fBD). ORB fea-
298 tures [27] extracted inside Ω_{static} from the first frame
299 are matched against each late-window frame under
300 robust geometric filtering; fBD ($\downarrow$) is the median
301 matched-point displacement, averaged over the late
302 window and expressed as a percentage of the frame
303 diagonal. It captures accumulated geometric displace-
304 ment of persistent structure and is intentionally insen-
305 sitive to fine-scale flicker that moves no structure (for-
306 mal definition in Supp. S1). 307

308 Normalized Background Flow (NBF). Complement-
309 ing accumulated drift, NBF ($\downarrow$) is the mean optical-
310 flow magnitude inside the static mask, averaged over
311 all frame pairs and normalized by frame width for res-
312 olution invariance and by the inter-frame interval Δt
313 for frame-rate invariance, giving units of 10^{-3} frame
314 widths per second (Supp. S1). Here Δt is the inter-
315 val of the sampled pairs actually used for flow esti-
316 mation: methods are evaluated at their native frame
317 rates, which differ across the roster, and the evalua-
318 tion pipeline subsamples each clip to a common tem-
319 poral rate before measurement. Expressing NBF per
320 unit time rather than per frame makes the quantity
321 independent of both the native rate and the sampling
322 policy, so the measure remains comparable if either
323 changes. NBF is a normalized magnitude, not a ratio:
324 dividing by whole-frame flow would couple background
325 leakage to foreground behavior and inflate the measure
326 for methods whose dynamic motion decays. 326

4.2. Flow Persistence

327 Global-motion compensation. Coherent scene dis-
328 placement is removed before dynamic-region motion is
329 evaluated. A single translation vector is insufficient:
330 under rotation or zoom the static-region flow field is
331 not constant, and its component-wise median is near
332 zero for a centered rotation, so a translation-only esti-
333 mate would leave exactly the corruptions Sec. 5 injects
334 uncompensated. We therefore estimate a robust global
335 similarity transform $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$ from static-
336 region correspondences under RANSAC, and subtract
337 the displacement field it induces from the measured
338 flow pointwise, giving a compensated field $\tilde{\mathbf{u}}_t$ (Supp.
339 S1). A similarity rather than an unrestricted affine
340 model is used because affine shear absorbs legitimate
341 local deformation and over-corrects. Where correspon-
342 dences are insufficient we fall back to median transla-
343 tion and record the fallback per sequence. 344

345 Motion-Compensated Foreground Flow (MCFF).
346 MCFF ($\uparrow$) is the mean compensated flow magnitude

inside Ω_{flow} over a temporal window. We report it on both the early and the late window—MCFF-E and MCFF-L—so a method with uniformly negligible motion cannot appear strong through relative measures alone (Supp. S1).

Flow Persistence (FP). FP ($\uparrow$) is the ratio of late to early MCFF, clipped at a fixed constant to prevent unstable ratios when early motion is small (Supp. S1). FP near one indicates retained motion; substantially smaller values indicate decay; values above one are not automatically better. FP is never reported without its MCFF magnitudes.

4.3. Drift Leakage and Attenuation

Whole-frame metrics cannot determine whether apparent dynamic-region motion arises from intended flow or from displacement of the whole scene; we report two diagnostics, since neither subsumes the other. Let F_{static} denote mean flow magnitude over Ω_{static} , and F_{raw} , F_{comp} the mean raw and compensated magnitude over Ω_{flow} , all in the late window, where drift has accumulated.

Drift Leakage Ratio (DLR). DLR ($\downarrow$), the ratio of static-region to raw dynamic-region flow magnitude in the late window, locates the sequence’s motion energy (Supp. S1). It is not a fraction and is deliberately unbounded: $\text{DLR} > 1$ indicates that static support moves more than the region intended to move, a regime that occurs in practice.

Drift Attenuation Ratio (DAR). DAR ($\downarrow$), one minus the ratio of compensated to raw dynamic-region flow, measures the signed relative attenuation of measured dynamic-region flow after global similarity compensation (Supp. S1). Reported values are clipped to $[0, 1]$ while the signed quantity is retained for validation; negative values, which arise where local flow opposes the estimated global field, are reported rather than suppressed.

Interpretation. Neither is a causal decomposition of motion into drift and non-drift components. DLR and DAR are secondary diagnostics that divide the work between them: DLR is a translational-leakage diagnostic whose rotational response is inverted, so rotational drift is carried by fBD and DAR instead, and that division is why DLR is never read alone. DAR is signed, and its negative values are real. High DLR with moderate DAR indicates static-region motion a rigid model does not account for; it does not identify what the residual is.

Semantic guardrail. The decomposition is agnostic to what moves: a sequence rendering drifting fog where a river was requested can occupy a favorable operating point. The natural guardrail is a prompt-alignment score—CLIP-based image-text similarity [8], or a learned video-quality predictor [7] at higher cost. We do not report one, and the factors should be read accordingly: they establish where motion occurs and whether it persists, not that the depicted phenomenon is the one requested. Pairing SNF-Bench with a semantic score is how it should be used in practice.

4.4. Joint Interpretation

No factor is read alone: low drift with low compensated motion indicates an over-stabilized sequence, high compensated motion with high static-region flow indicates motion contaminated by displacement, and high retention with negligible magnitude indicates persistent but trivial motion. SNF-Bench therefore reports no single scalar score, since collapsing the factors would recreate the ambiguity it exists to expose.

Implementation convention. Videos are normalized to a common evaluation resolution and subsampled to a common flow-estimation rate; no frame interpolation is performed, since synthesizing frames would alter the flow field being measured. Every estimator, threshold, window extent and constant is fixed globally before the sweep and applied unchanged to every system, and the per-sequence frame rate, resolution and compensation mode are released alongside the scores.

5. Controlled Validation

Before SNF-Bench is used to compare generators, we validate that its measurements respond to the failure modes they target. A benchmark is not justified merely because its metrics rank differently; each must change selectively, monotonically, and in the expected direction under known corruptions. We therefore construct controlled variants of fixed-camera reference clips with known corruption type and severity, independently of the audit in Sec. 6: perturbation strengths, masks, windows, and hyperparameters are fixed before the final results are inspected.

5.1. Perturbations and Criteria

Given a reference sequence we generate variants that isolate individual components of the decomposition. Geometric drift applies a temporally increasing global transformation—translation, rotation, or scale—simulating camera-like background displace-

ment that accumulates as long-horizon error does. Motion attenuation progressively blends dynamic-region content toward an early reference, and late freezing replaces all content after a chosen fraction of the sequence, both leaving static support untouched. Photometric drift varies brightness and color temperature while preserving geometry and motion, bounding the residual sensitivity of flow-based quantities. Temporal repetition replaces late dynamic content with a short repeated cycle. Partition perturbation erodes and dilates the region boundary rather than the video.

Geometric severity is specified as the mean pixel displacement a corruption induces, with each family’s parameter derived from it using the frame geometry. This is not presentational: a parameter in degrees is not commensurate with one in pixels, and at the evaluation resolution one degree of rotation induces only a few pixels of displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude. A factor is admitted only if its response direction matches its definition, that response is monotonic in severity, unrelated corruptions produce substantially smaller changes, and conclusions survive perturbation of the region partition. Factors failing these checks are reported as diagnostics rather than headline quantities. The severity specification and the admission outcome for every factor are in the supplementary material.

Complementary checks. Mechanistic validation shows a factor responds to the corruption it names, not that its ordering matches what an observer perceives, and we do not close that gap. The supplementary material specifies an axis-specific protocol—forced choice on one failure axis, never on which output is better—and reports the one instance we ran: an open-weight judge [26] failed a presentation-order control, so no agreement rate is drawn from it. Perceptual agreement remains open.

5.2. Validation Results

Figure 2 and Table 1 report the outcome. Injected translation produces the expected monotone increase in both static-fidelity factors, and progressive late freezing drives MCFF-L and FP down—to 0.02 and 0.07 of their unperturbed values once the late window is frozen—while leaving the static factors comparatively unchanged, which is the selectivity the decomposition requires. At maximum translation, fBD and NBF reach $1.86\times$ and $1.32\times$ baseline, while VBench Dynamic Degree [11] reaches only $1.07\times$ and improves on a corruption fixed-camera evaluation should penalize.

Two results qualify the framework rather than sup-

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	translational leakage	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	–

Table 1. Mechanistic validation and factor admission. ρ is signed target-family Spearman correlation; ρ_{rot} reports rotational response; off-target is the largest relative excursion under photometric and mask perturbations. Admission requires $\rho \geq 0.7$ and off-target response below 25%. Rotation is reported but not gated: DLR is inverted there, while fBD and DAR carry that case. VBench DD is a contrast row, not a candidate factor. Twelve clips per severity.

port it. Rotation is the harder geometric case for every factor, since correspondences degrade and the induced displacement is smallest at the frame center: both static-fidelity factors respond more weakly than under translation (+0.70 for fBD, +0.60 for NBF, against +1.00 for both), so neither is used alone. DLR is the sharper limitation—admitted on its target family, where it responds perfectly, but with an inverted rotational response (−0.80), so it is read only as a comparative diagnostic alongside DAR. Reported alongside these is the scope boundary of Sec. 5.1: looping content sustains both persistence and magnitude, so SNF-Bench cannot distinguish genuine progression from repetition, and does not claim to.

6. Public-Model Audit

We apply SNF-Bench to publicly released long-horizon generators. The goal is not a universal leaderboard but a determination of how checkpoints occupy the static-fidelity/flow-persistence space under one recorded common T2V configuration.

Systems and setting. The audit covers 7 publicly released autoregressive or streaming generators with accessible checkpoints, frozen as a set before the sweep. Every T2V checkpoint is loaded into the same recorded inference implementation rather than its released pipeline. Fixed before the sweep to give the distilled checkpoint family a shared step budget, the configuration uses a four-step denoising schedule (indices warped by the released scheduler), six frames per block, seed 0, and one rollout/cache path. The text-conditioned runs also share 832×480 output at 16 fps. Within a horizon, checkpoint weights and their recorded regular/EMA selection are the only system-specific inference inputs. Accordingly, these results

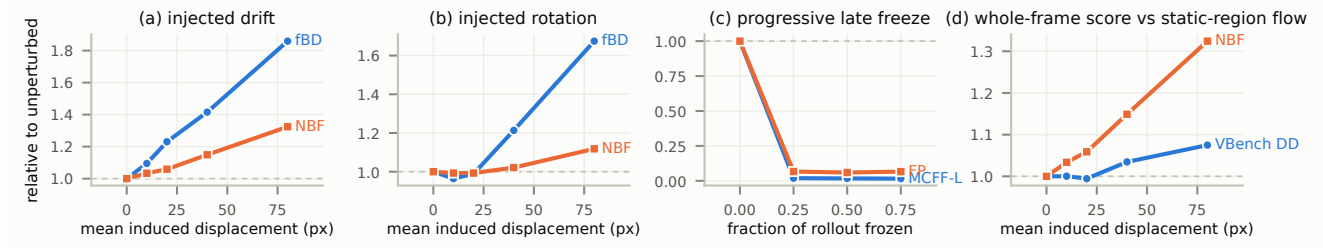


Figure 2. Mechanistic validation. Curves are relative to unperturbed fixed-camera clips. Translation raises fBD and NBF; progressive late freezing collapses MCFF-L and FP; under the same translation, VBench Dynamic Degree changes little and slightly upward. The perturbations are synthetic and implicate no published method.

characterize checkpoints under this common configuration, not each method’s published inference procedure; a checkpoint distilled for another step budget is not observed at its own operating point. The I2V track instead separates released-pipeline outputs from dagger-marked long-horizon-wrapper rows (Sec. 7.3).

Controls and statistics. Within each item every system receives identical conditioning, and the seed is fixed before the sweep. This makes each run reproducible but does not make different checkpoints’ stochastic trajectories equivalent; cross-system comparison rests on the paired-item design. Videos are retained in their recorded form and the normalization of Sec. 4 applied unchanged. Scores are computed per item and averaged over items; the sensitivity of every conclusion to category macro-averaging is reported in the supplementary material. Uncertainty is estimated by percentile bootstrap [4] over items, and we report 95% marginal intervals. These intervals summarize per-checkpoint uncertainty; all headline separation claims use paired bootstrap distributions of per-prompt differences, reported in Supplementary Table S3. Measurement coverage and fBD abstentions are reported in Supplementary Tables S6–S7; incomplete metric records are counted as partial rather than silently treated as successful. To test whether the decomposition changes interpretation, the same outputs are additionally scored with established whole-frame metrics and the induced orderings compared. The focus is interpretation change, not a claim that existing metrics are wrong.

7. Results and Interpretation Changes

Unless stated otherwise, T2V comparisons refer to checkpoints under the recorded common configuration. I2V released-pipeline and wrapper results are identified separately and never ranked across settings.

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 23$)				
CausVid	8.08 ± 3.15	9.27 ± 4.14	$2.87 \rightarrow 1.96$	0.67 ± 0.18
Self-Forcing	11.4 ± 3.63	12.7 ± 3.97	$10.4 \rightarrow 2.54$	0.44 ± 0.22
Infinite-Forcing	7.75 ± 3.84	3.65 ± 0.88	$1.17 \rightarrow 0.29$	0.63 ± 0.22
Rolling-Forcing	16.8 ± 3.69	11.2 ± 2.70	$3.18 \rightarrow 1.57$	0.75 ± 0.29
Reward-Forcing	14.8 ± 3.71	8.51 ± 2.79	$3.81 \rightarrow 0.80$	0.49 ± 0.18
LongLive	15.9 ± 2.89	13.9 ± 4.76	$6.46 \rightarrow 2.41$	0.75 ± 0.28
Causal-Forcing	22.4 ± 2.26	87.6 ± 18.6	$9.84 \rightarrow 7.21$	0.83 ± 0.31
120s horizon ($n = 6$)				
CausVid	6.15	3.42	$0.36 \rightarrow 0.52$	1.06
Self-Forcing	21.2	6.80	$4.85 \rightarrow 0.82$	0.85
Infinite-Forcing	16.1	2.60	$0.50 \rightarrow 0.32$	0.64
Rolling-Forcing	23.8	10.6	$3.33 \rightarrow 0.77$	0.38
Reward-Forcing	20.0	11.6	$2.05 \rightarrow 0.69$	0.39
LongLive	22.5	9.66	$5.10 \rightarrow 1.19$	0.81
Causal-Forcing	21.8	130	$10.5 \rightarrow 10.7$	0.90

Table 2. Recorded common-configuration T2V audit. A four-step denoising schedule with released-scheduler warping, six frames/block, and seed 0. n is common within each horizon. Means $\pm$ half a percentile-bootstrap 95% interval (10k resamples, seed 0); $n < 10$ shows means only. MCFF-E→L accompanies FP. Separation uses paired per-prompt intervals (Table S3). DLR/DAR are supplementary.

7.1. Recorded Common-Configuration T2V Audit

Table 2 reports the primary 60s T2V tier and a 120s directional check; Figure 3 visualizes one 60s prompt. Because the 120s panel has 6 different prompts, it is not compared column-by-column with 60s. The 5 and 240s diagnostics and all four horizons are supplementary.

Motion persistence. We read (MCFF($\mathcal{L}$), FP) jointly: sustained flow needs non-trivial late motion and high retention. MCFF-E beside MCFF-L exposes uniformly negligible motion.

Operating regimes. Joint static-fidelity and persistence columns expose low-drift/low-motion, high-motion/high-drift, and low-drift/non-trivial-motion regimes without a scalar score. Infinite-Forcing [12]



Figure 3. Every audited text-conditioned system on one prompt. Rows show $t = 0/30/60$ s; columns are ordered left-to-right by increasing static-region drift (fBD). The two highlighted systems are the pair a whole-frame motion score cannot separate.

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 30$)				
Causal-Forcing++ (2-step) [†]	9.18	5.97	0.85 → 0.60	0.86
Causal-Forcing++ (1-step) [†]	5.34	7.05	0.86 → 0.57	0.77
Causal-Forcing (frame-wise) [†]	21.1	36.4	2.80 → 2.69	0.89
Causal-Forcing++ (2-step, frame-wise)	9.94	6.91	1.30 → 0.78	1.02
Self-Forcing [†]	17.3	5.52	1.50 → 0.69	0.89
CausVid [†]	19.9	3.31	0.58 → 0.41	0.81
Causal-Forcing++ (1-step, frame-wise)	10.4	5.25	0.53 → 0.52	1.01

Table 3. I2V released pipelines and deployment sensitivity. Unmarked rows are released-pipeline outputs; [†] rows use the wrapper; settings are not compared. n is common within each horizon. Means; intervals are supplementary. MCFF-E→L accompanies FP. No cross-setting claim. DLR/DAR are supplementary.

reduces measured dynamic-region flow by that proportion; it is attenuation, not causal decomposition. The residual is unidentified and may include non-rigid instability, intended motion, partition contamination, or flow error.

7.3. Image-Conditioned (I2V) Track: Released Pipelines and Deployment Sensitivity

At 60 s, Table 3 separates two released-pipeline outputs from five dagger-marked wrapper rows; it is not a seven-system audit, and settings are never ranked. The panel shows deployment sensitivity without attributing its cause. Unlike T2V, each system shares one source image, so static support is given rather than imagined differently. Complete horizons and 5 s bidirectional references are supplementary.

has the lowest NBF under the recorded common configuration—paired per-prompt intervals versus all 6 peers exclude zero (Supp. Table S3)—but also the least surviving late motion. Causal-Forcing [43] occupies the opposite, higher-background-flow corner.

7.2. Whole-Frame Motion and Static-Region Flow Yield Different Interpretations

Across 7 checkpoint means, Dynamic Degree tracks static-region flow at $\rho = +0.93$ (descriptive; per-item analysis in the supplementary material): more apparent motion also tends to mean more background motion. Causal-Forcing ranks 1/7 on Dynamic Degree but 7/7 on NBF; Infinite-Forcing mirrors it at 1/7 and 7/7. Causal-Forcing’s DLR of 0.94 indicates static-region motion energy nearly equal to dynamic-region energy. Its DAR of 0.40 means fitted similarity compensation

8. Limitations and Conclusion

SNF-Bench is limited to long-horizon fixed-camera generation where static support and dynamic flow separate. It measures neither physical nor semantic realism, cannot distinguish progression from repetition, and depends on an automatic partition bounded by our erosion–dilation study.

Whole-frame motion and region-resolved static fidelity can materially change the interpretation of the same outputs. Reporting static fidelity and flow persistence separately, with drift quantities only as context, complements broad benchmarks without a composite score. The released specification and common-configuration checkpoint audit provide a reproducible basis for evaluation; independent partitions and broader non-water coverage remain important next steps.

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